

Personalization & Authenticity Setup

Each student must begin their notebook with the following block (given in the template). This creates a **unique signature and random seed** based on your name, roll number, and cohort salt.

```
# ==== Cohort Binder (DO NOT MODIFY) ====
import hashlib, json, random, numpy as np, datetime, os, matplotlib.pyplot as plt

STUDENT_NAME = input("Enter your full name: ").strip()
ROLL_NO      = input("Enter your roll no: ").strip()
SALT         = "AIML_Week3_2025"    # fixed by instructor

sig_str = json.dumps({"n":STUDENT_NAME, "r":ROLL_NO, "s":SALT})
STUDENT_SIG = hashlib.sha256(sig_str.encode()).hexdigest()[:12]
SEED = int(STUDENT_SIG[:8], 16) % (2**31-1)
random.seed(SEED); np.random.seed(SEED)

TODAY = datetime.date.today().isoformat()
OUT_PREFIX = f"AIML_{STUDENT_SIG}_{TODAY}"
os.makedirs("out", exist_ok=True)

def stamp_plot(title):
    plt.title(f"{title}\nSIG:{STUDENT_SIG}  DATE:{TODAY}")

print(f" Student: {STUDENT_NAME} | Roll: {ROLL_NO}")
print(f" Signature: {STUDENT_SIG} | Seed: {SEED}")
print(f" Output prefix: out/{OUT_PREFIX}_")
```

All subsequent plots, figures, and saved models must include the **signature** and **date watermark**.

This ensures every notebook is verifiably original.

Task Structure

- The project consists of **five tasks** in total.
- **Tasks 1–3 are compulsory** and must be **properly written, executed, and produce valid outputs**.
- Each must include your **unique signature and random seed logic** as provided in the template.
- Tasks should contain:
 - Code implementation
 - Output (plots/metrics/tables)
 - Short reflection (2–3 lines)

Mandatory Tasks

1. **Classification Challenge** – Logistic Regression & Decision Tree
2. **Regression Challenge** – Linear Regression on continuous data
3. **Clustering Challenge** – Unsupervised learning using K-Means

Optional Tasks (Advanced)

4. **Deep Learning (PyTorch)** – Build a simple NN or CNN
5. **NLP (Transformers)** – Sentiment analysis using Hugging Face

Only Tasks **1–3** are required for successful completion. Any use of GPTs will be easily identified and if found you will be deregistered from the programme

Tasks 4 and 5 are optional, but completing them can move you to a **higher or deeper level of the problem statement**.

Task 1: Classification Challenge (Logistic Regression & Decision Tree)

Goal: Predict a categorical outcome using your own dataset.

Requirements:

1. Select any dataset (minimum 150 rows, 4+ features) with a *binary or multi-class target* (e.g., Pass/Fail, Disease/No Disease, Iris dataset, etc.).
2. Split data into train/test using `random_state = SEED`.
3. Build and compare:
 - Logistic Regression
 - Decision Tree (with `max_depth = 2 + (SEED % 5)`)
4. Evaluate using:
 - Accuracy, Precision, Recall, F1-score
 - Confusion Matrix (saved as image with signature)
5. Write a short reflection (3–4 lines):

“Which model performed better and why?”

“What feature influenced the result most?”

Deliverables:

- Table of metrics
 - Confusion matrix plot
 - Written reflection
-

Task 2: Regression Challenge (Linear Regression)

Goal: Predict a continuous variable from a small dataset.

Requirements:

1. Choose a dataset where target is numeric (e.g., house prices, student marks, sales).
2. Drop one feature based on your signature:
3. `drop_col_idx = SEED % X.shape[1]`
4. `X = X.drop(X.columns[drop_col_idx], axis=1)`
5. Train a **Linear Regression** model.
6. Evaluate with MAE, MSE, RMSE, and R²-score.
7. Plot *Actual vs Predicted* values with red diagonal line.
8. Write 3–4 lines:

“Which variables contribute most?”

“Was it underfitting or overfitting?”

Deliverables:

- Metric values
 - Scatter plot (with signature watermark)
 - Reflection
-

Task 3: Clustering Challenge (K-Means)

Goal: Discover natural patterns in your dataset.

Requirements:

1. Select any dataset (≥ 3 numeric features).
2. Use **Elbow Method** to determine k (or $k = 2 + (\text{SEED} \% 3)$).
3. Perform **KMeans** clustering and visualize in 2D using PCA.
4. Create a contingency table (true labels vs clusters) if labels exist.
5. Reflect on:

“Do clusters align with reality?”

“What did you learn about hidden structure in data?”

Deliverables:

- Elbow curve

- PCA scatter plot
 - Reflection (3–4 lines)
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Task 4: Deep Learning Mini-Project (PyTorch Neural Network)

Goal: Train a simple **Neural Network or CNN** to classify images.

Requirements:

1. Use **MNIST, Fashion-MNIST**, or your own 3-class image dataset.
 - If personal dataset: capture 15–30 photos of 3 objects (e.g., pen, notebook, phone).
 - Include 1 image with a handwritten label “AIML Week-3 <signature> <today>”.
2. Architecture must use:
3. `hidden_units = 64 + 16 * (SEED % 5)`
4. Train for **3–5 epochs** using Adam optimizer & CrossEntropyLoss.
5. Report accuracy and show a **3×3 prediction grid** (correct vs wrong).
6. Reflection (3–4 lines):

“Which classes confuse the model?”

“What could improve accuracy?”

Deliverables:

- Training logs & accuracy
 - Grid of predictions
 - Reflection
-

Task 5: NLP with Transformers (Hugging Face Sentiment Analysis)

Goal: Apply a pre-trained transformer to real text.

Requirements:

1. Write **10–15 original sentences** about your learning experience (half positive, half negative).
 - Must be **written by you**, not copied or generated.
2. Use Hugging Face pipeline("sentiment-analysis") to analyze.
3. Display a clean table of results: sentence → label → score.
4. Reflect:

“Did the model interpret tone correctly?”
“Where did it fail?”

Deliverables:

- Table of predictions
- Reflection